Context-Aware Alt Text Support

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Selected Barriers

Missing alt text when browsing the web:

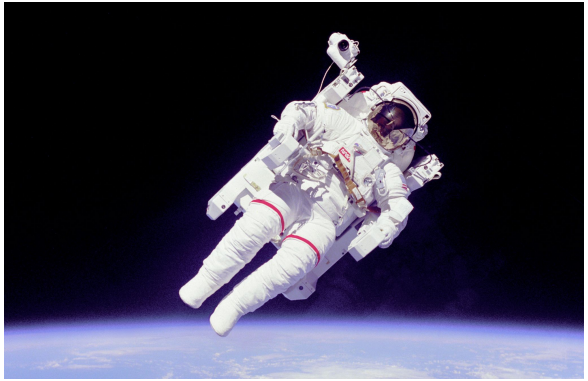
- Forgotten by authors
- Decorative images are mixed with informational images

Many websites contain images without alt text causing problems for:

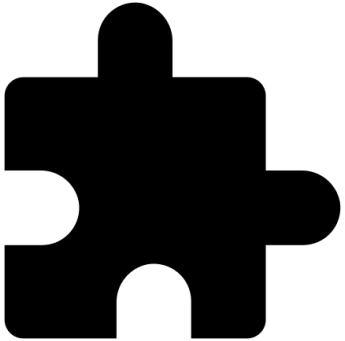
- Screen reader users
- Low vision users
- Users with cognitive disabilities

Resulting Issues:

- Screen readers aren’t able to describe images
- Users will miss important information and lose context
- An unequal access to content is created



Approach to Addressing the Barriers



Solution: Accessibility Browser extension - Alt Text generator

Uses:

- Identify inaccessible images by detect images missing alt text.
- Generate AI-based alt text by using the context of the surrounding text and the image content.
- Allow users to customize descriptions to their needs.
- Distinguish between decorative and informative text

Goal: To make inaccessible websites more accessible by filling in missing alt text

Design

Process:

1. Extension scans webpage images
2. Identifies missing alt text attributes
3. AI generates description based on surrounding context
4. Injects alt text dynamically, allowing for user customization

Features:

- Accessibility warning icon
- Generated alt text preview
- Accept/Reject button
- Accessibility score indicator

How the Design Promotes Equity

Our solution helps:

- Visually impaired users
- Low vision users
- Screen reader users
- Older users with vision decline

Equity benefit:

Provides equal access to visual information at the same time as other users. This supports individual browsing instead of relying on outside sources for aid.

Prototype & Plan

Week 1: Detect images and isolate those with missing alt text.

Week 2: Integrate AI caption generator using the image and surrounding page context.

Week 3: Create UI pop up screen for user input and feedback on the alt text.

Week 4: Testing prototype on example sites and fix bugs or broken functionality.

Prototype Goal: Show a working prototype on a limited selection of web pages.